



Massachusetts Institute of Technology

Freight Transportation Analysis

SCM.C51 Machine Learning

Today's Agenda

01

RESEARCH RECAP

Reviewing the research question, data scope, EDA, and LightGBM performance.

02

LIGHT GBM VS TRANSFORMERS INITIAL RESULTS

Comparison of old model to new predicted model that can better capture the information.

03

DEEP DIVE INTO EXOGENOUS DATA

Integrating macroeconomics signals and external features to capture broader market context.

04

CONCLUSION

Analyzing attention mechanisms and quantifying feature value.

Can Transformer based models **capture nonlinear temporal dependencies and market dynamics** to improve lane-level freight rate forecasts compared to baseline models?

Our Research Question

Why Transformers?

Freight pricing is spatially distributed and highly variable, reflecting dynamic, time-dependent behavior rather than simple static relationships.

1

Non-Stationarity & Regime Shifts

- 2021 super-cycle caused structural breaks in pricing behavior
- Mean and variance shift over time → violates assumptions of ARIMA / linear models
- Transformers adapt via time-dependent representations

2

Long-Range Temporal Dependencies

- Freight rates influenced by long-horizon effects (weeks–months)
- Linear models rely on short memory (lag-based)
- TFT uses attention to learn dependencies across entire encoder window (~26 weeks)

3

Multi-Variable Interactions

- Price driven by interacting factors:
 - Lagged prices
 - Supply/demand signals
 - Calendar effects
 - Macro variables
- Transformers capture nonlinear cross-feature relationships via attention

4

Probabilistic Forecasting Under Uncertainty

- Business decisions require P10 / P50 / P90 forecasts
- Not just point predictions
- TFT outputs full predictive distributions across multiple horizons



Data Selection

■ Data selection process for high volume lane identification and filtering

Equipment Type ~

Restricted to Van Only (220,795 rows, ~88% of total) to ensure a consistent pricing regime and avoid noise from specialized equipment.

Volume Threshold ~

Lanes ranked by total quote count to ensure sufficient panel observations for robust time-series modeling and sequence learning.

Gap Rate Filtering ~

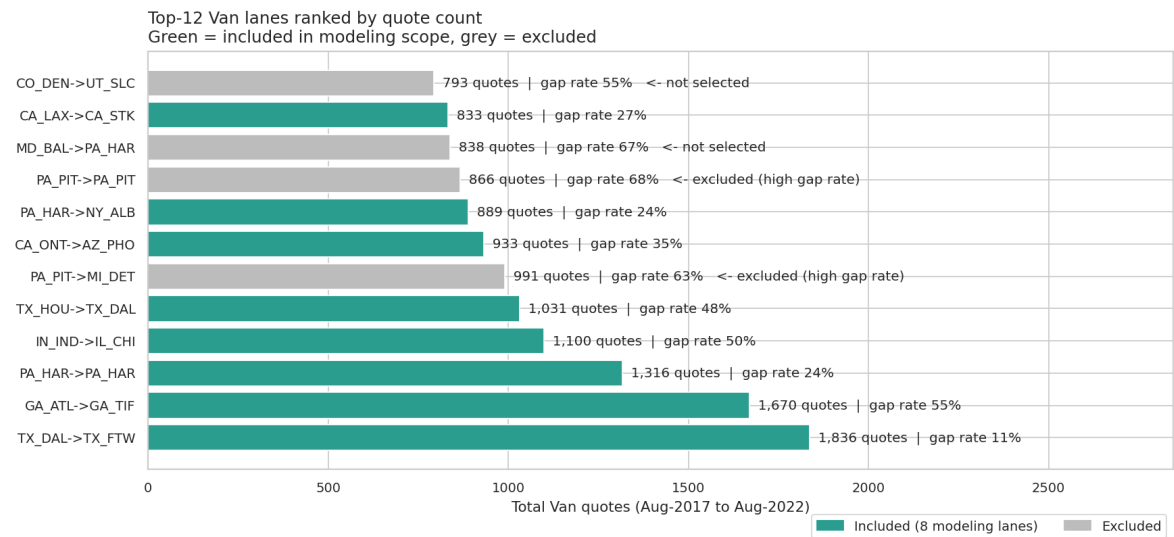
Dropped lanes with a weekly gap rate above 50%. Sparse lanes break sequence models and inflate training noise.

8 MODELING LANES SELECTED

TX_DAL	→	TX_FTW
GA_ATL	→	GA_TIF
PA_HAR	→	PA_HAR
IN_IND	→	IL_CHI
TX_HOU	→	TX_DAL
CA_ONT	→	AZ_PHO
PA_HAR	→	NY_ALB
CA_LAX	→	CA_STK

Lane-Level Freight Forecasting Framework

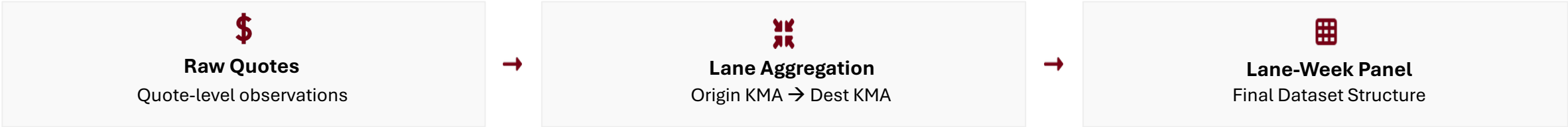
Transitioning from exploratory analysis to setting data up for modelling



LANE (ORIGIN -> DEST)	TOTAL QUOTES	WEEKS ACTIVE	GAP RATE	MEDIAN PRICE	MEDIAN MILEAGE	QUOTES/WEEK
TX_DAL -> TX_FTW	1,836	302	11%	\$450	131 mi	3
GA_ATL -> GA_TIF	1,670	339	55%	\$814	250 mi	0
PA_HAR -> PA_HAR	1,316	201	24%	\$625	61 mi	6
IN_IND -> IL_CHI	1,100	305	50%	\$552	214 mi	1
TX_HOU -> TX_DAL	1,031	337	48%	\$750	261 mi	1
CA_ONT -> AZ_PHO	933	343	35%	\$1,250	316 mi	2
PA_HAR -> NY_ALB	889	176	24%	\$1,078	287 mi	4
CA_LAX -> CA_STK	833	308	27%	\$1,200	329 mi	2

Feature Engineering at Lane (KMA) Level

Transforming raw quote-level data into structured time-series inputs for forecasting models (56 total modelling features selected)



lane	week	horizon	price	lag_1	lag_4	roll_4_mean	quote_count	matched_carriers	mean_weight	week_sin	month_cos	mileage	origin_lat	dest_lat
TX_DAL→TX_FTW	2021-W10	2	500	480	460	470	7	12.5	39,500	0.87	0.50	131	32.8	32.7
CA_ONT→AZ_PHO	2021-W10	4	1250	1200	1100	1150	3	9.2	42,100	0.87	0.50	316	34.0	33.4

Each row represents one (lane × week) observation with engineered features

**Identifiers**

lane, horizon: Informs the model exactly where the lane is and how far ahead it needs to forecast.

**Lagged & Rolling**

price_lag_X, price_lag_4, price_roll_*:
Captures market memory and autoregressive dynamics.

**Supply & Demand**

quote_count, matched_carriers_mean:
Real-time proxy for market tightness and pricing pressure.

**Weekly Aggregates**

price, mean_price, std_price: Captures the full distribution and dispersion of prices within a week.

**Calendar Encoding**

week_sin, month_cos: Encodes cyclical seasonality without assuming fixed, rigid patterns.

**Static Lane Features**

mileage, lat/long, urbanicity: Encodes fixed structural differences across the network geography.

How do baseline and transformer models perform when forecasting freight spot prices using **only historical target data**, before incorporating any exogenous variables?

Modeling Approach

Compare LightGBM model to establish baseline KPIs with advanced transformer models

We model freight pricing as a **multi-horizon forecasting problem**, predicting weekly lane-level price ranges (P10, P50, P90) using both **LightGBM (feature-based)** and **TFT (sequence-based)** models. Using **walk-forward validation**, we simulate real-world conditions and evaluate performance with metrics that capture both **accuracy and uncertainty**.

Problem Formulation

Objective

Forecast weekly median lane-level spot prices across multiple horizons: 1, 2, 4, and 8 weeks ahead.

Uncertainty Quantification

Quantile regression framework predicting P10, P50, and P90 values to capture market volatility and risk.

Validation Strategy

Walk-Forward Cross-Validation

F1: 2017–2019 → 2020

F2: →2020 → 2021

F3: →2021 → 2022



Prevents data leakage + captures structural regime shifts.

Modeling Framework

LightGBM (Gradient Boosted Trees)

Tabular, feature-driven approach with separate models per quantile to handle non-linear lane dynamics.

Temporal Fusion Transformer (TFT)

Attention-based sequence model that directly learns complex temporal dependencies and regime shifts.

Evaluation & Key Design Choices

Metrics

MAE (Point Accuracy), Quantile Loss (Primary), R^2 loss, and Coverage (P10–P90 Reliability).

Design Choices

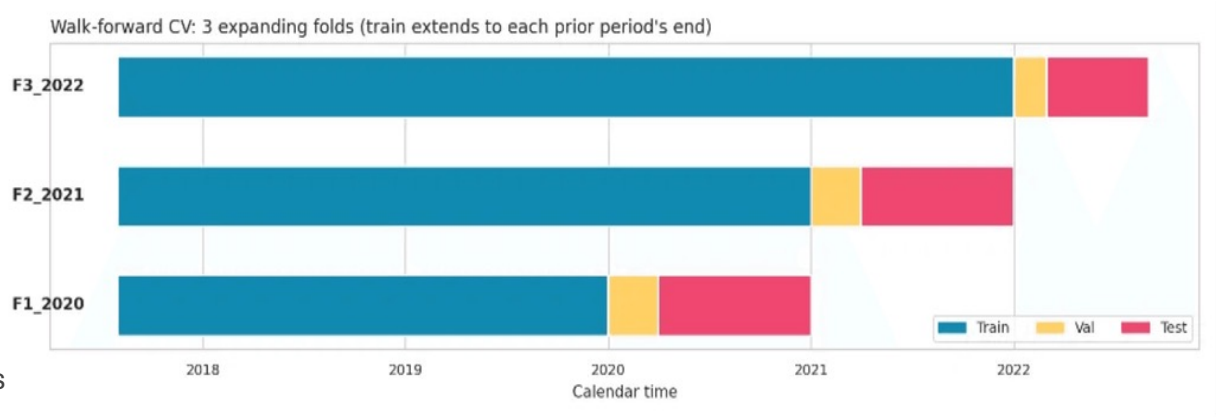
- Lagged features shifted to prevent leakage
- Continuous time indexing for missing weeks

LightGBM Modeling Strategy

Using Walk-Forward Validation & Framework Architecture

WHY THIS APPROACH?

- ✔ **Non-Stationary Market Dynamics:** Freight markets undergo **structural shifts** (e.g., **COVID supply chain shocks**). Standard validation fails to account for these regime changes, leading to poor generalization.
- ✔ **Temporal Realism & Leakage:** In production, we only have historical data. Walk-forward CV strictly respects the time arrow, ensuring the model never "sees" the future during training or validation.
- ✔ **Expanding Window Memory:** By expanding the training set over time, we allow the model to **learn both long-term seasonal priors and recent market volatility** across multiple economic cycles.



FOLD	TRAINING PERIOD	VALIDATION (EARLY-STOPPING)	TEST PERIOD (OUT-OF-TIME)	ROWS
F1_2020	2017-08 ... 2019-12-31	2020-01-01 ... 2020-03-31	2020-04-01 ... 2020-12-31	1,134
F2_2021	2017-08 ... 2020-12-31	2021-01-01 ... 2021-03-31	2021-04-01 ... 2021-12-31	966
F3_2022	2017-08 ... 2021-12-31	2022-01-01 ... 2022-02-28	2022-03-01 ... 2022-08-31	612

FRAMEWORK: LIGHTGBM

A high-performance Gradient-Boosted Decision Tree (GBDT) model. It can capture complex, non-linear interactions in tabular time-series data while maintaining computational speed.

MODELING VARIANTS

- v1: Raw price baseline (Default)
- v2: Log-transformed target (Stability)
- v3: Tuned log model (Optimized)
- v4: CQR calibration (Uncertainty)

EVALUATION METRICS

Multi-dimensional performance tracking:

- MAE
- MAPE
- Quantile Loss
- Coverage

Focusing on both point accuracy and interval reliability.

LightGBM Feature Ablation and Results

Feature ablation across 56 lagged inputs shows which signals consistently improve forecast accuracy and which introduce noise.

What was Tested

A1: Naive

Price lag only (baseline)

A2: Calendar

Time features (month, week)

A3: Static

Distance & Geography

A4: Lagged & Rolling

Multi-lag prices and trends

A5: + Supply (Full Modes)

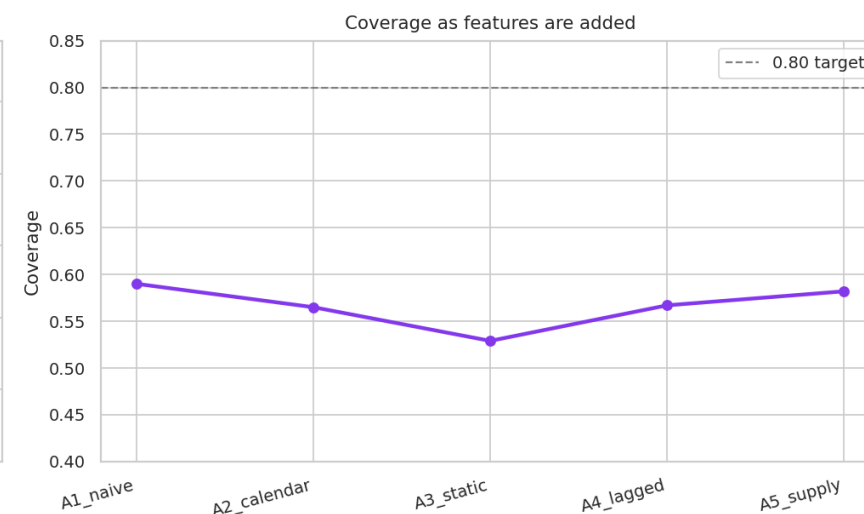
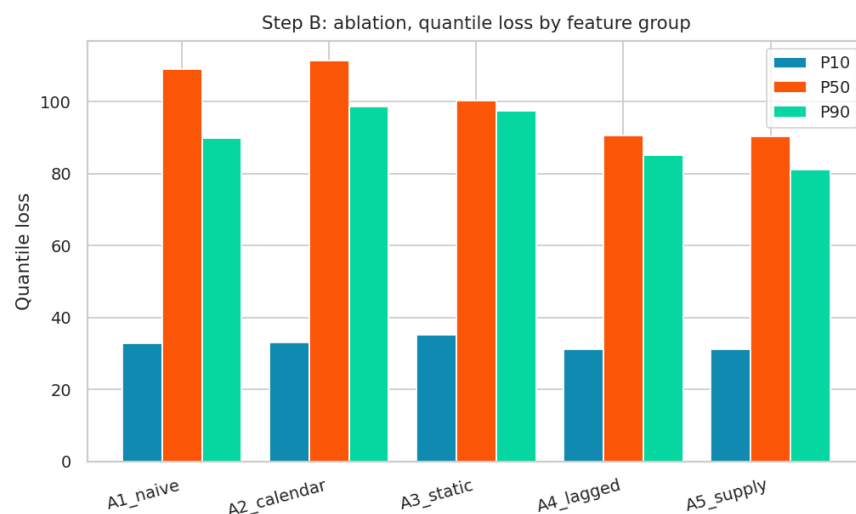
HQ Locations, Within-week load, etc.

KEY RESULTS

Biggest performance gain: Lagged + Rolling feature reduced error by approximately 11%. It captures market memory and serves as the dominant signal in forecasting.

Calendar Features Hurt: Adding calendar features alone increases error by ~5%.

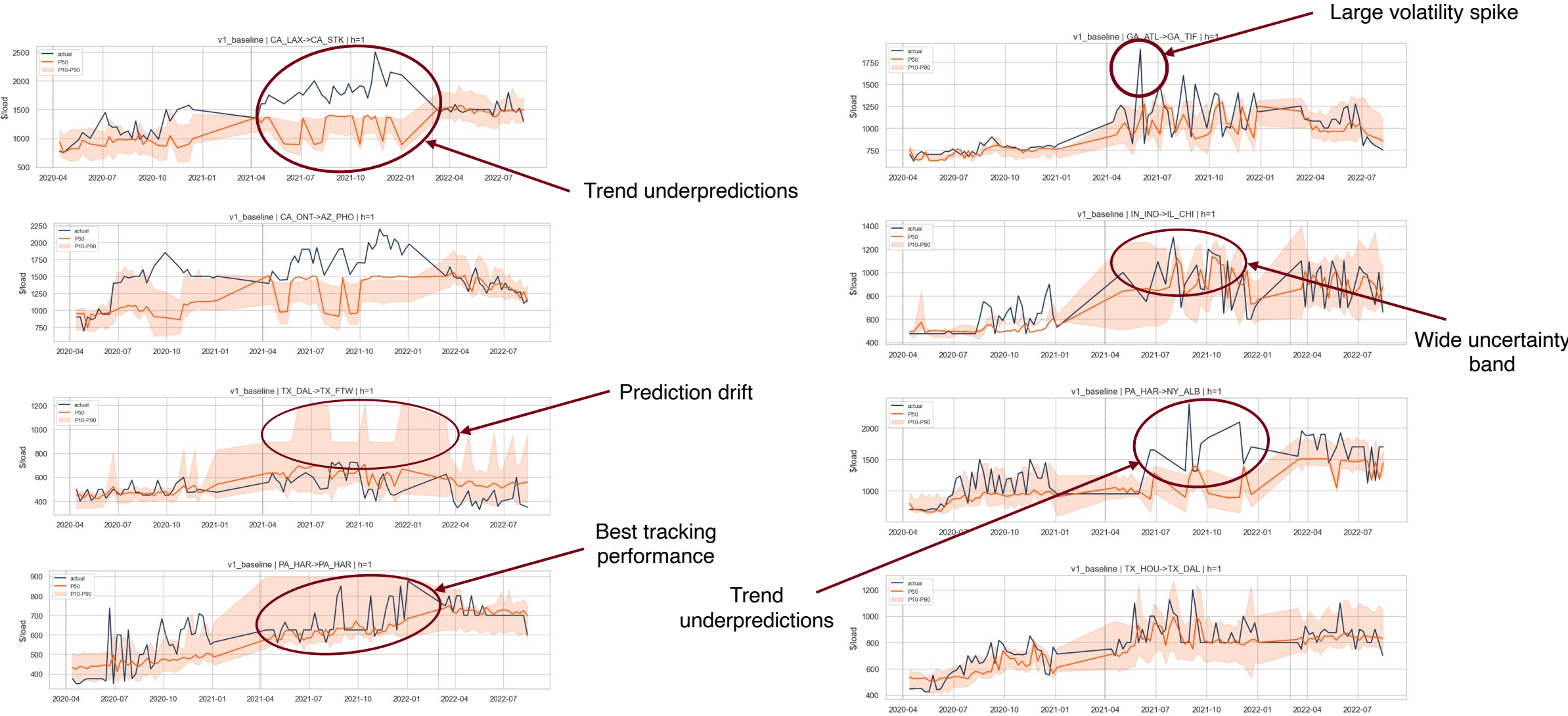
Supply Features: Modest ~2% average gain, but critical during 2021 market spikes to capture real-time market tightness.



Feature ablation shows that lagged price and rolling features are the dominant drivers of performance, reducing error by **~11%**. Calendar features alone actually **hurt performance**, while supply-demand features provide small average.

LightGBM 8-Lane Results

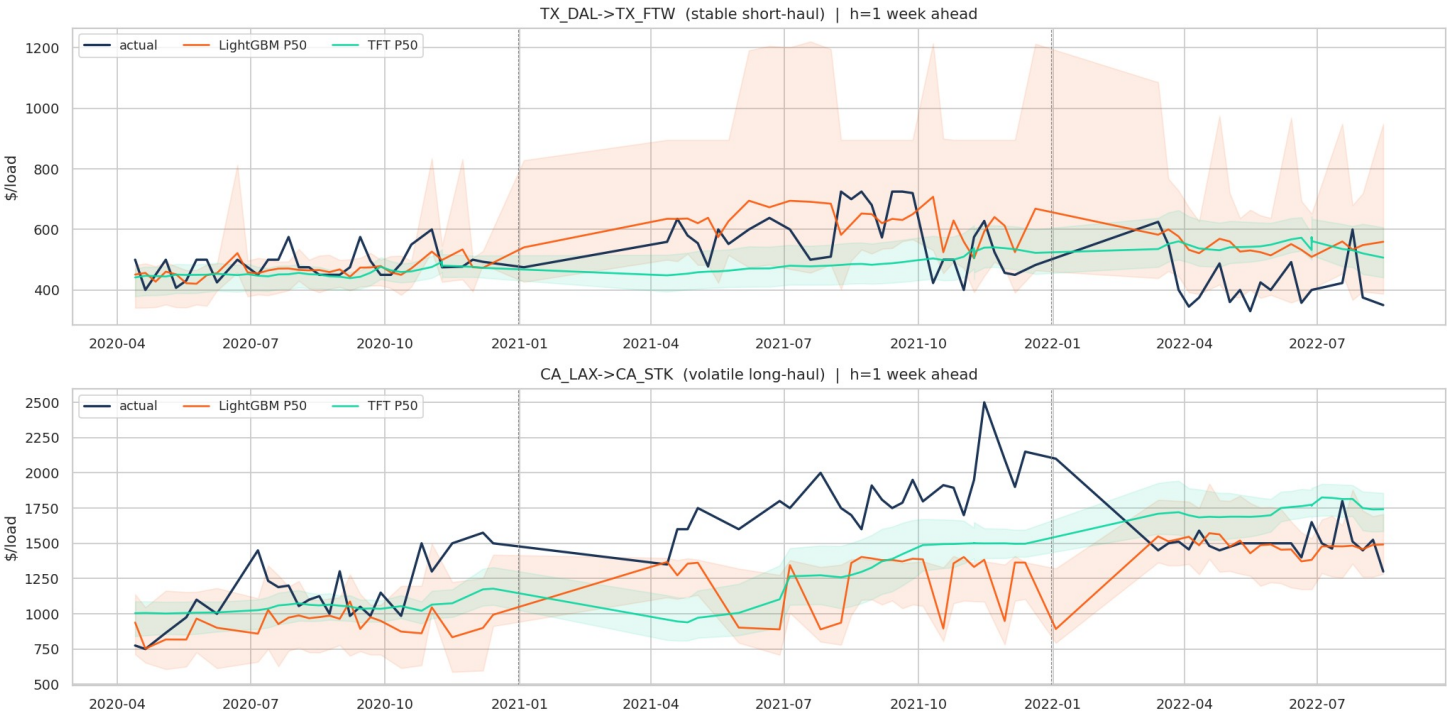
Compare LightGBM model to establish baseline KPIs with advanced transformer models



Transformers (TFT) Architecture & Results

Using Walk-Forward Validation & Framework Architecture

Hyperparameter	Value
max_encoder_length	26 weeks
max_prediction_length	8 weeks
hidden_size	16
attention_head_size	2
dropout	0.10
target_normalizer	GroupNormalizer(log)
loss	QuantileLoss([0.1, 0.5, 0.9])
learning_rate	1e-3



Summary of Model Comparison Results

Compare LightGBM model to establish baseline KPIs with advanced transformer models

1. Data Regime

- ~1,500 observations → small dataset
- Tree-based models outperform deep learning in low-data settings

2. Feature Engineering

- Advantage Lag features, rolling stats, calendar already encoded
- LightGBM leverages engineered features efficiently
- TFT’s ability to learn these patterns becomes redundant

3. Quantile Modeling

- LightGBM trains separate models per quantile
- TFT shares parameters → weaker performance on tails (P10/P90)

Key Results

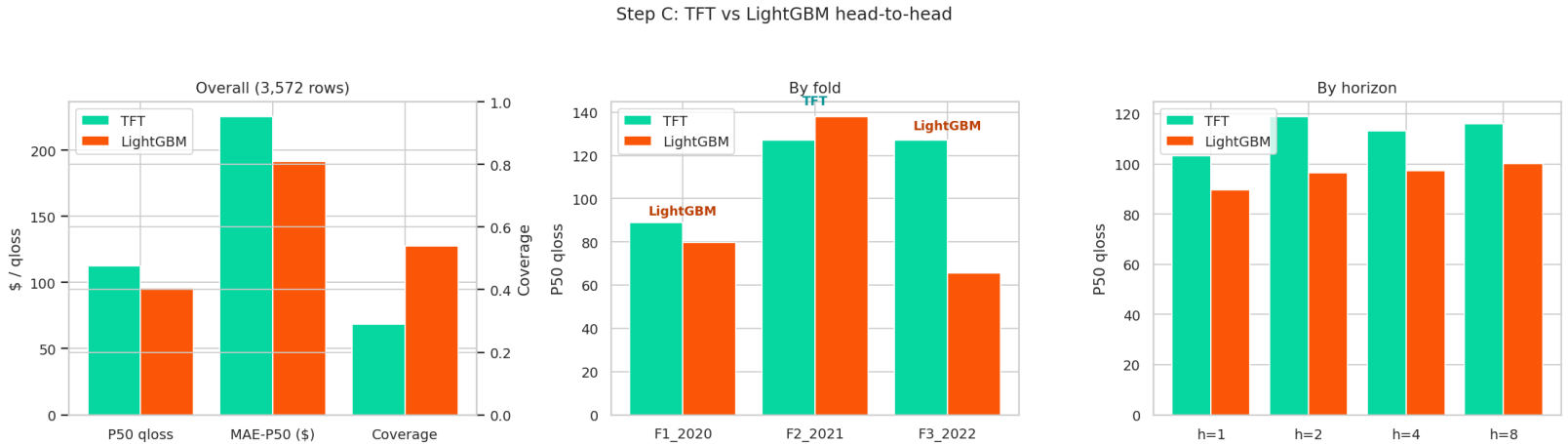
Across 96 (fold × lane × horizon) evaluations:

- LightGBM wins: 66 (69%)
- TFT wins: 30 (31%)

LightGBM Pros

- Lower MAE (191 vs 225)
- Lower P50 loss (95.7 vs 112.60)
- Better coverage calibration

model	qloss_q10	qloss_q50	qloss_q90	mae_p50	mape_p50	coverage_10_90
TFT	59.11	112.58	89.81	225.6	22.27	0.29
LightGBM	32.41	95.72	87.21	191.43	16.81	0.54



Feature Importance

Compare LightGBM model to establish baseline KPIs with advanced transformer models

Objective: evaluate baseline model performance using only historical data and establish a lower-bound benchmark for comparison with exogenous enhanced models

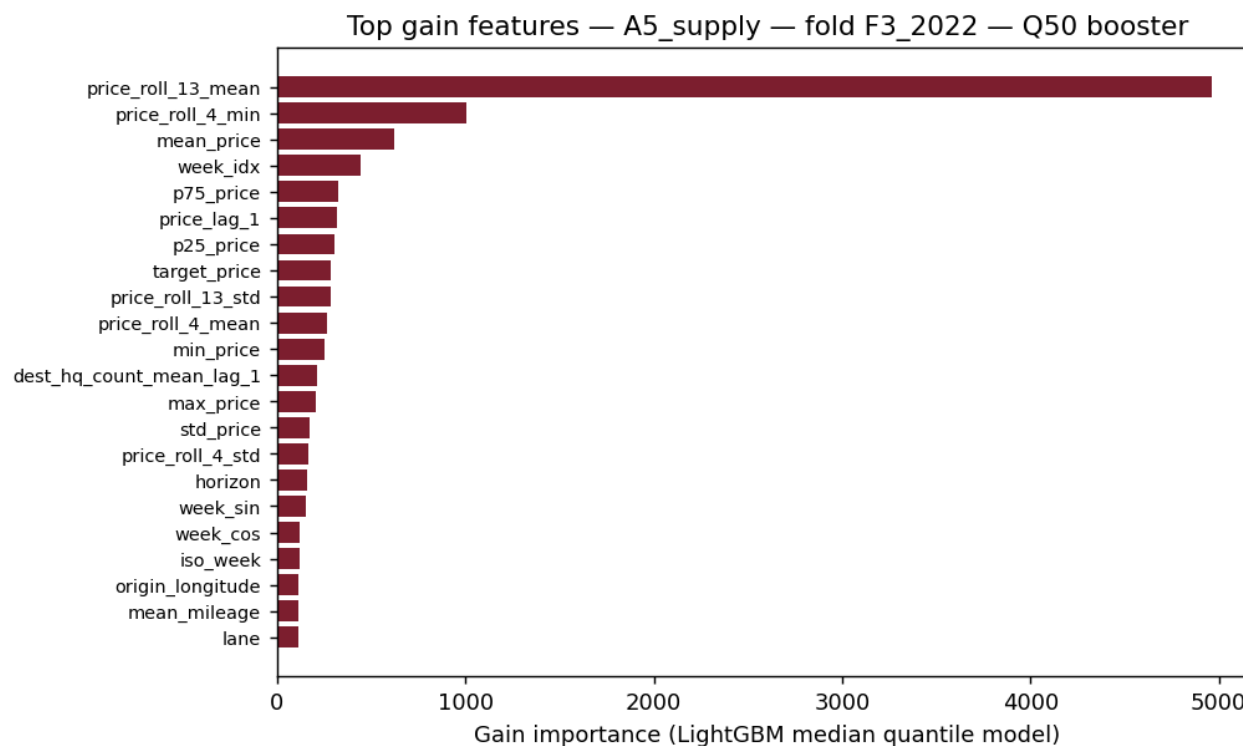
Model Setup

- Inputs: Historical time-series features only
- No external drivers (e.g., fuel price, demand signals, macro factors)
- Models tested:
 - LightGBM (baseline)
 - Temporal models (TFT / sequence models)



Results:

- Performance degradation observed without exogenous inputs
- Increased prediction error in a higher MAE/RMSE
- Reduced ability to capture demand shock, market volatility, and non-stationary trends

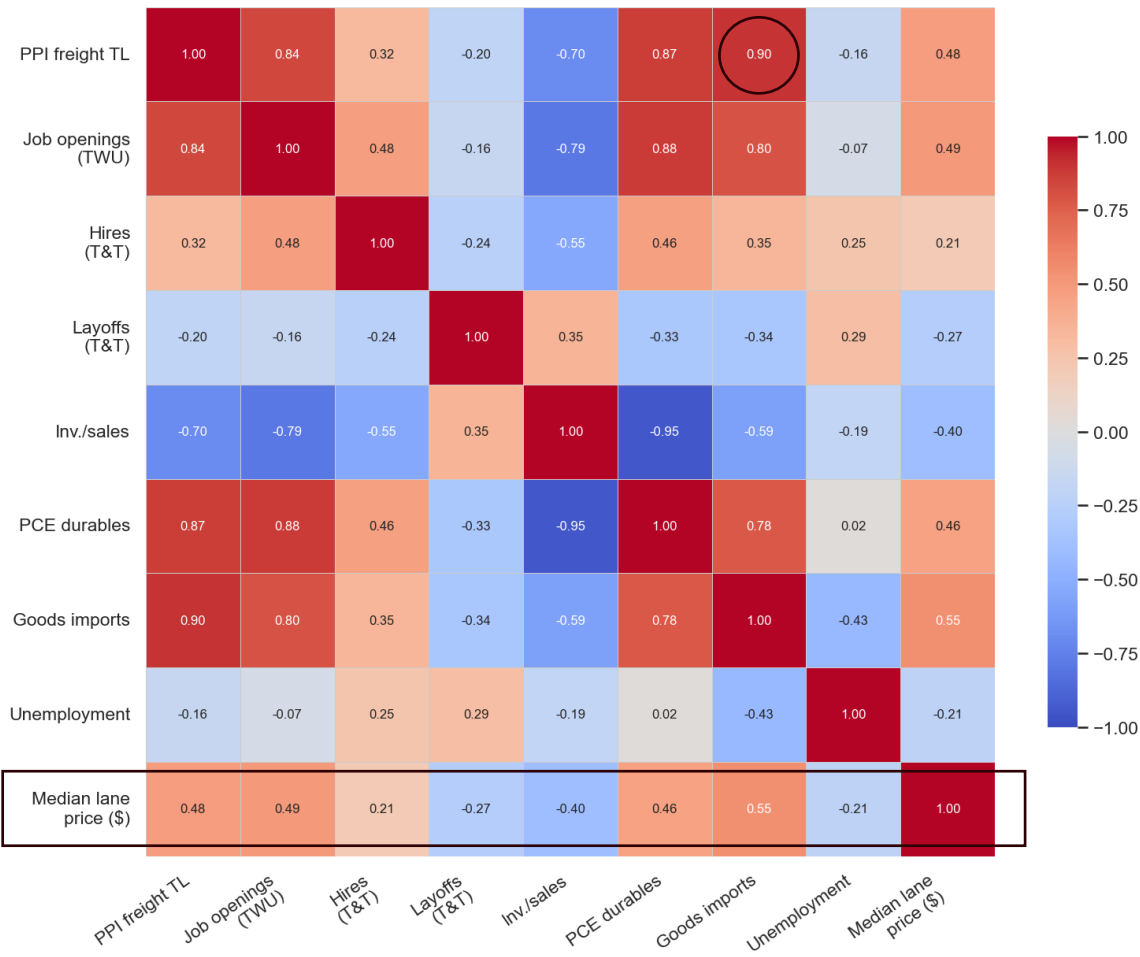


How does model performance change when **exogenous market, temporal, and lane-level features** are introduced into both the baseline and transformer forecasting frameworks?

Exogenous (Macro) Variables

Price vs. Macro Variables correlation

Pearson correlation — eight lagged exogenous series + cross-lane median weekly price
(weekly grain; macros carry 4-week publication lag as in panel)



Highly correlated exogenous variables may contain redundant information, so variables such as **Goods Imports** and **Producer Price Index: Freight** could be removed because they have over 90% correlation with features like Job Openings and PCE Durables

TOP CORRELATIONS (Pearson, 1 month lag to current price)

- +0.90** Goods Import with PPI Freight Goods import volumes strongly move with freight demand
- +0.88** PCE Durables and Job Openings
- +0.87** Producer Price Index: Freight TL with PCE Durables
- +0.84** PPI Freight TL with Job Opening
- 0.95** Inventory to Sales Ratio with PCE Durables
- 0.79** Inventory to Sales Ratio with Job Openings

Exogenous Variables vs Price

Our Hypothesis

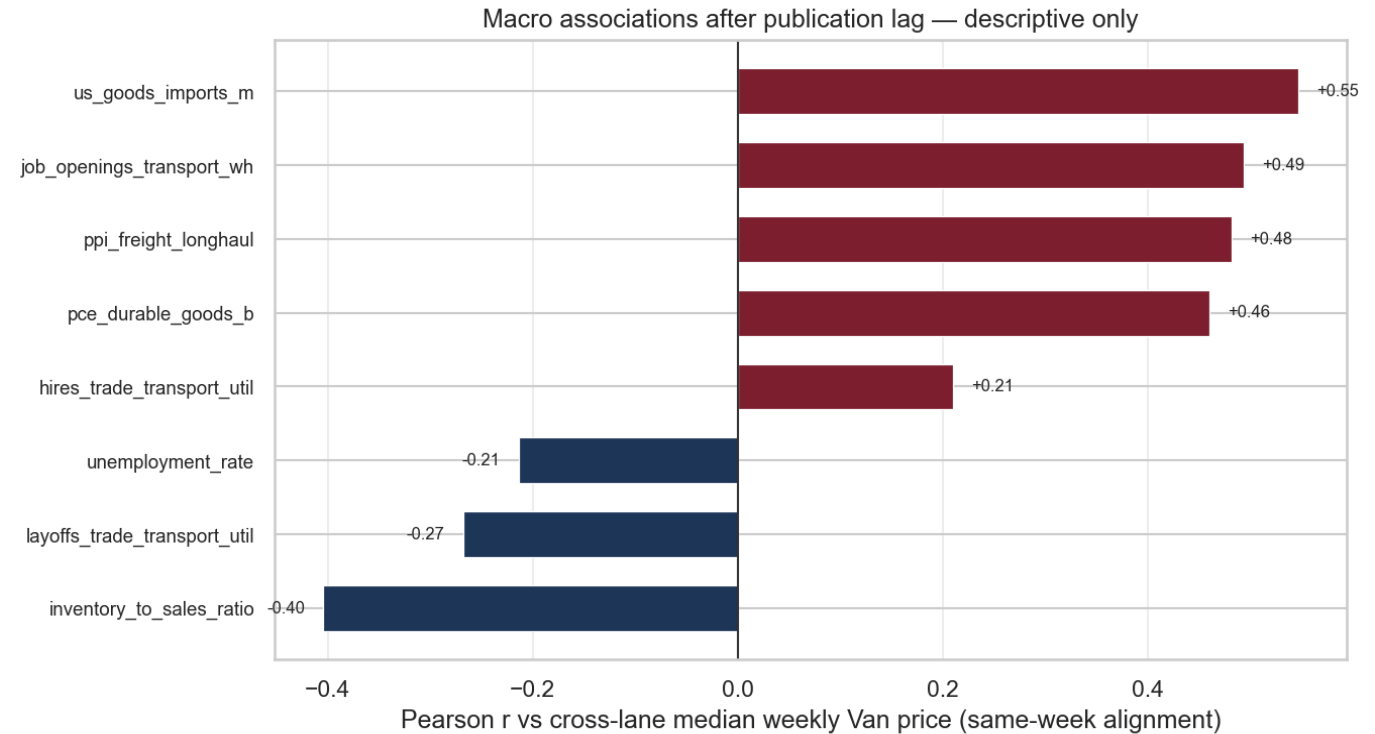
The higher the correlation when lagged will determine which exogenous data to focus on and which to remove.

Key Insights

- Import activity and transportation job openings show the strongest positive relationship with freight prices.
- Inventory-to-sales ratio is negatively correlated, suggesting excess inventory suppresses shipping demand.
- Correlation-based feature selection helps reduce noisy exogenous inputs before transformer training.

Future Work

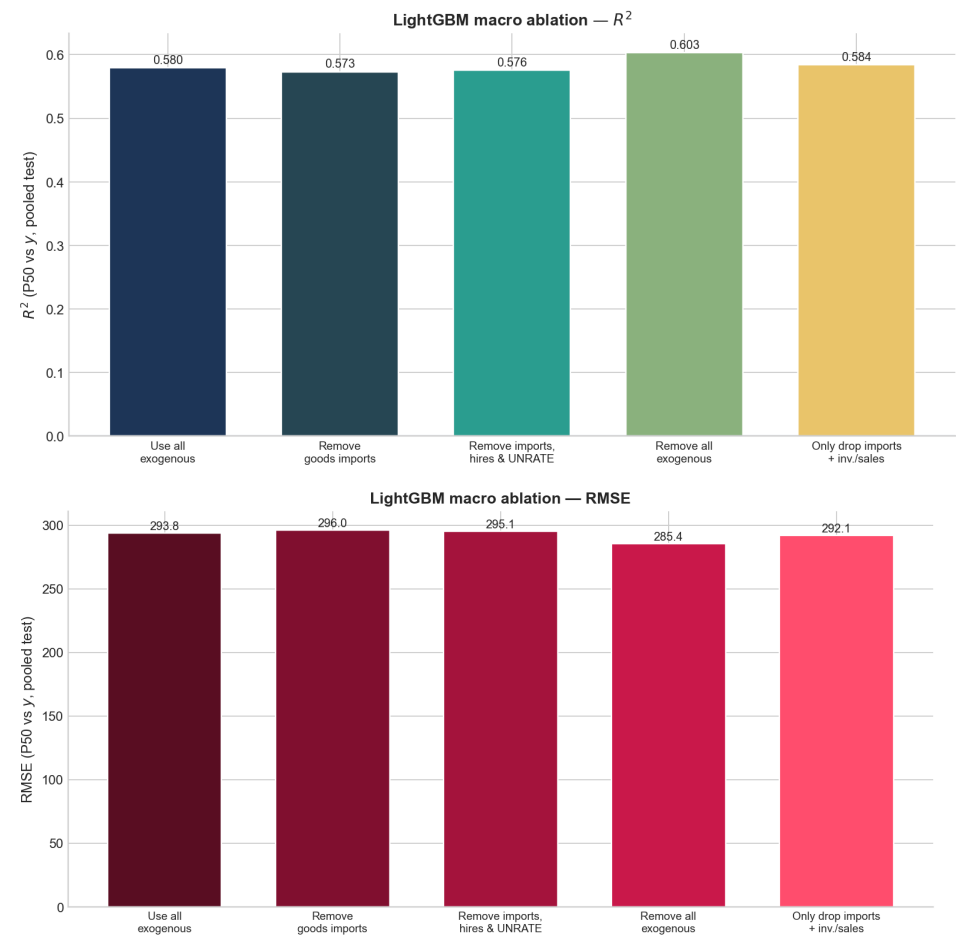
- Run the correlation matrix with different lag times and consider different exogenous data with highest correlation based on different lag times.



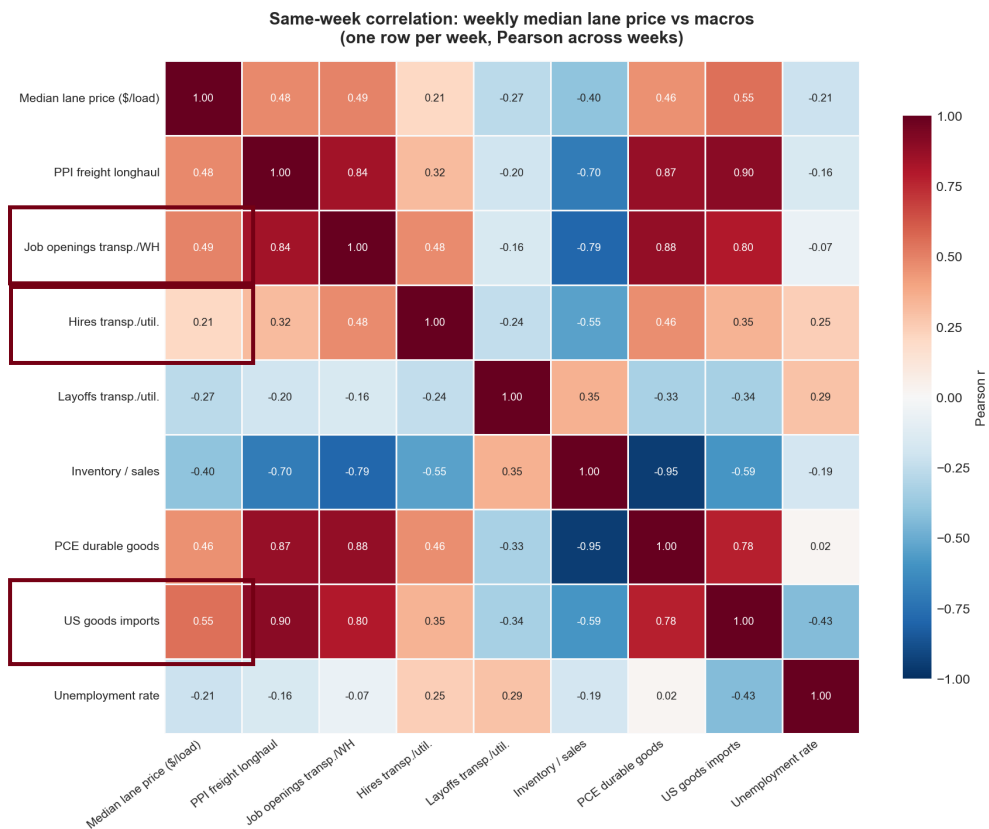
Note: All the exogenous data are lagged by one month and checked for correlation with the current price.

Exogenous Feature Impact

Price vs. Macro Variables correlation



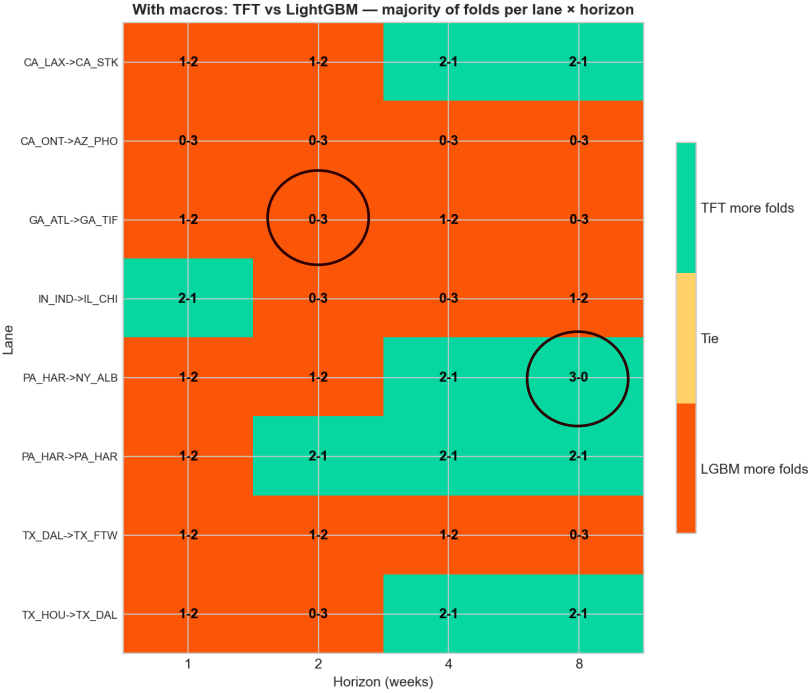
Stepwise ablation of exogenous variables shows negligible impact on model performance, with R^2 and RMSE remaining largely unchanged across configurations.



Our Hypothesis

Lagged price (t-4) may capture most of the signal contained in exogenous variables due to strong correlation with underlying market and lane dynamics.

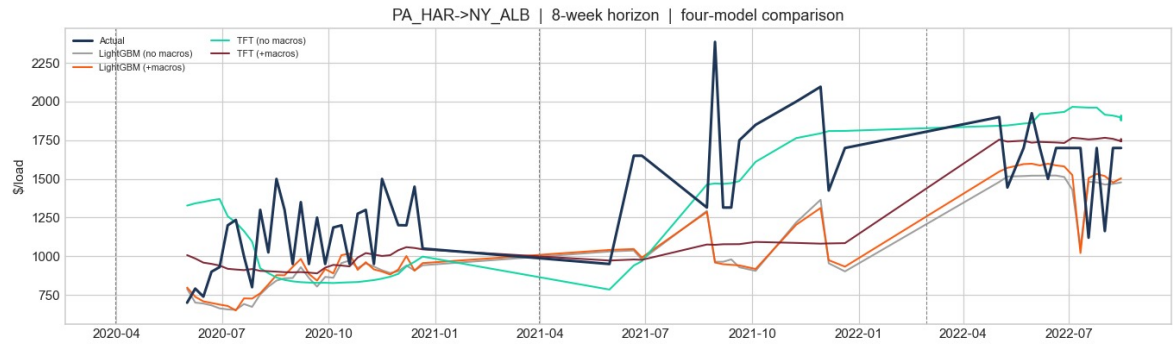
Results



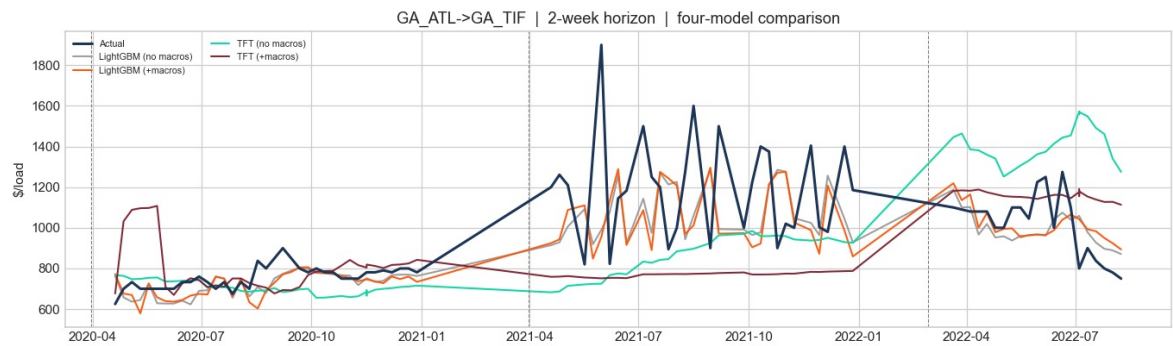
LightGBM for short horizons: LightGBM excels at 2-week, high-volatility forecasts because recent lag features are highly predictive and tree splits can react quickly to noisy short-term spikes without smoothing bias.

TFT for long horizons: Temporal Fusion Transformers excel at 8-week structural forecasts because attention mechanisms learn longer-range patterns and broader regime shifts when recent lags become less informative.

TFT Dominates



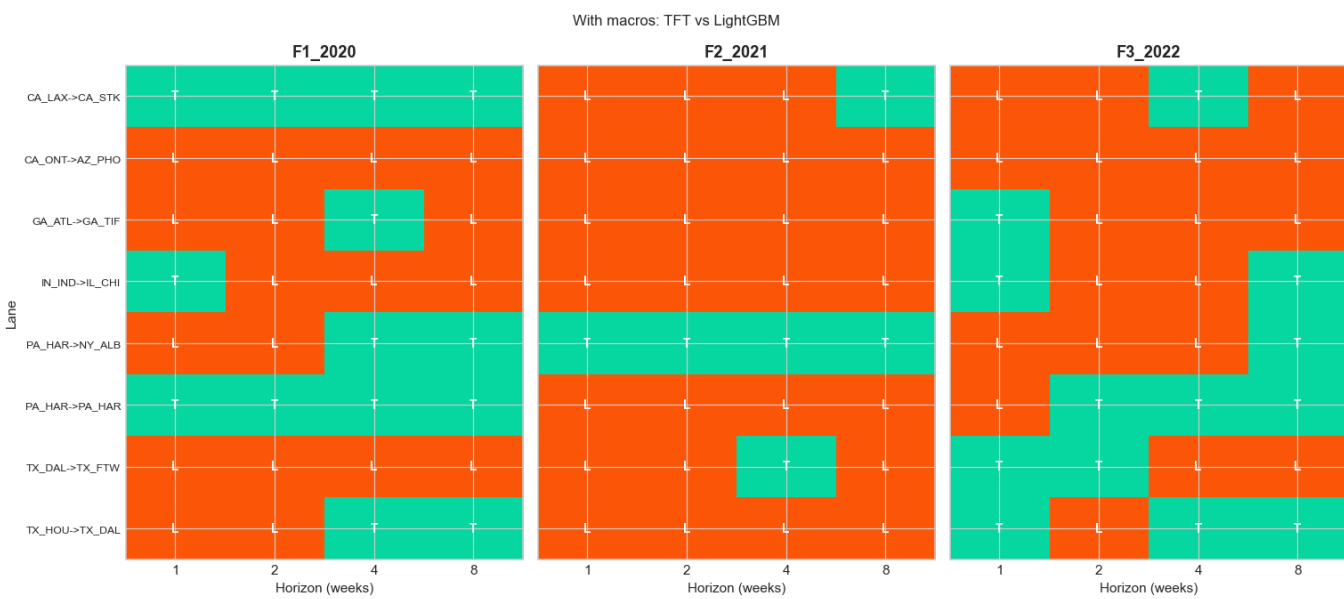
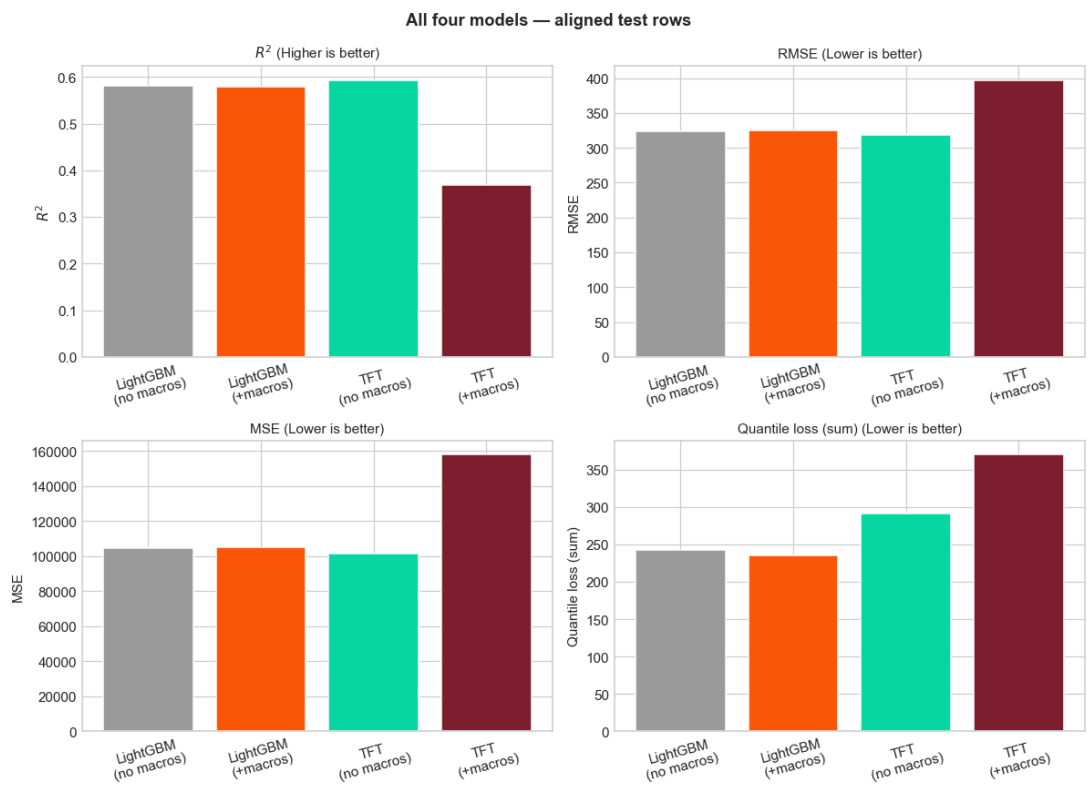
LightGBM Dominates



TFT results with exogenous

Price vs. Macro Variables correlation

- **Performance Impact:** R^2 shows slight improvement, but MSE/RMSE degrade due to sensitivity to noisy exogenous inputs.
- **Heatmap Gains:** More green cells vs. no-macro case, especially in volatile lanes and long horizons.
- **Limitation:** Small dataset (~1,500 lane-weeks) prevents TFT from fully learning cross-feature attention effectively.



All Model Comparison

Price vs. Exogenous Data correlation

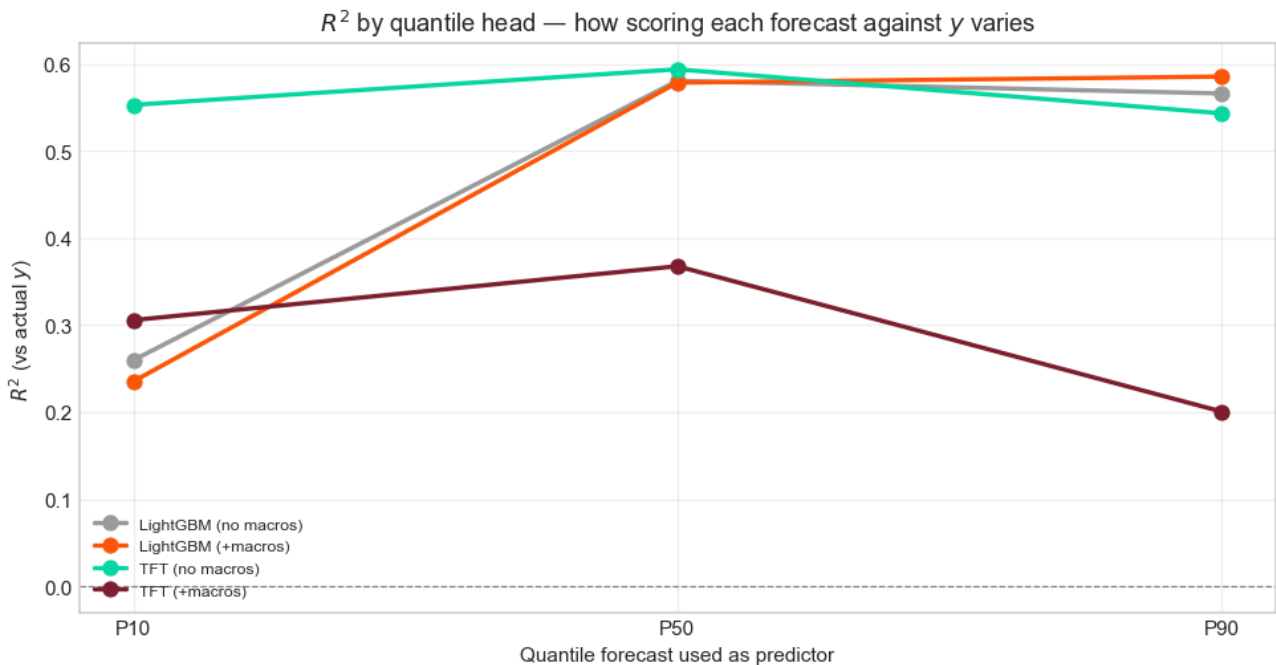
LightGBM

- Slight improvement in:
 - Tail performance (P90 R^2 $\uparrow$)
 - Coverage $\uparrow$
- Overall: Small but consistent gains

TFT

- Performance drops with macros
 - R^2 decreases significantly (especially P50, P90)
- Likely due to:
 - Noise / weak macro signal
 - Insufficient data to learn cross-feature attention

Model	r^2 p10	r^2 p50	r^2 p90	coverage_p10_p90
LightGBM	0.260	0.581	0.566	0.477
LightGBM (Exogenous)	0.235	0.579	0.586	0.542
TFT	0.553	0.594	0.543	0.221
TFT (Exogenous)	0.306	0.368	0.201	0.240



Model	F1_2020	F2_2021	F3_2022
LightGBM	0.547	0.418	0.821
LightGBM (Exogenous)	0.639	0.381	0.821
TFT	0.525	0.720	0.253
TFT (Exogenous)	0.576	0.189	0.395

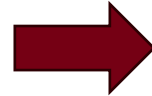
Conclusion and Takeaways

Recall: What did we set out to do and where we landed?



What did we set out to do?

- Weekly truckload-style panel: multi-horizon forecasts and uncertainty (quantiles / intervals), not only point accuracy.
- Compare a strong tabular baseline (LightGBM quantile) with a sequence model (TFT) suited to known vs unknown inputs.



Where we landed?

- Built a multi-horizon quantile forecasting model on a weekly lane panel with walk-forward evaluation.
- LightGBM delivered better point and quantile loss and much better 10–90 coverage than TFT in our main comparison.
- Macro features helped LightGBM; TFT v2 (with exogenous input) moved coverage in the right direction versus v1 (without exogenous input) but did not close the gap to LightGBM on the overall metrics we summarized.

Our Key Learnings

- Match the model to the lane activity, data shape, sample size, and the operational risk metric (not just average loss).
- Build a strong baselines first, quantile LightGBM is a serious production default for this problem class.
- Report coverage alongside loss, intervals that look tight can fail operationally or during the strategic sourcing stage.
- Transformers are not automatic wins, here they mainly illustrate when attention-based forecasting needs more evidence before replacing simpler tools.

Variation in model performance amongst different lanes



Which model wins which lane, fold and horizon?
(P50 Quantile Loss)

lane	F1_2020				F2_2021				F3_2022			
	1	2	4	8	1	2	4	8	1	2	4	8
TX_DAL->TX_FTW	TFT	TFT	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM
GA_ATL->GA_TIF	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM
PA_HAR->PA_HAR	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM
IN_IND->IL_CHI	TFT	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	LightGBM	TFT	LightGBM	TFT	LightGBM
TX_HOU->TX_DAL	LightGBM	LightGBM	LightGBM	TFT	LightGBM	LightGBM	LightGBM	TFT	LightGBM	LightGBM	LightGBM	LightGBM
CA_ONT->AZ_PHO	TFT	TFT	TFT	TFT	TFT	TFT	TFT	TFT	LightGBM	LightGBM	LightGBM	LightGBM
PA_HAR->NY_ALB	LightGBM	LightGBM	LightGBM	LightGBM	TFT	TFT	TFT	TFT	TFT	TFT	LightGBM	LightGBM
CA_LAX->CA_STK	TFT	TFT	TFT	TFT	TFT	TFT	TFT	TFT	LightGBM	LightGBM	TFT	LightGBM
Horizon (weeks)												

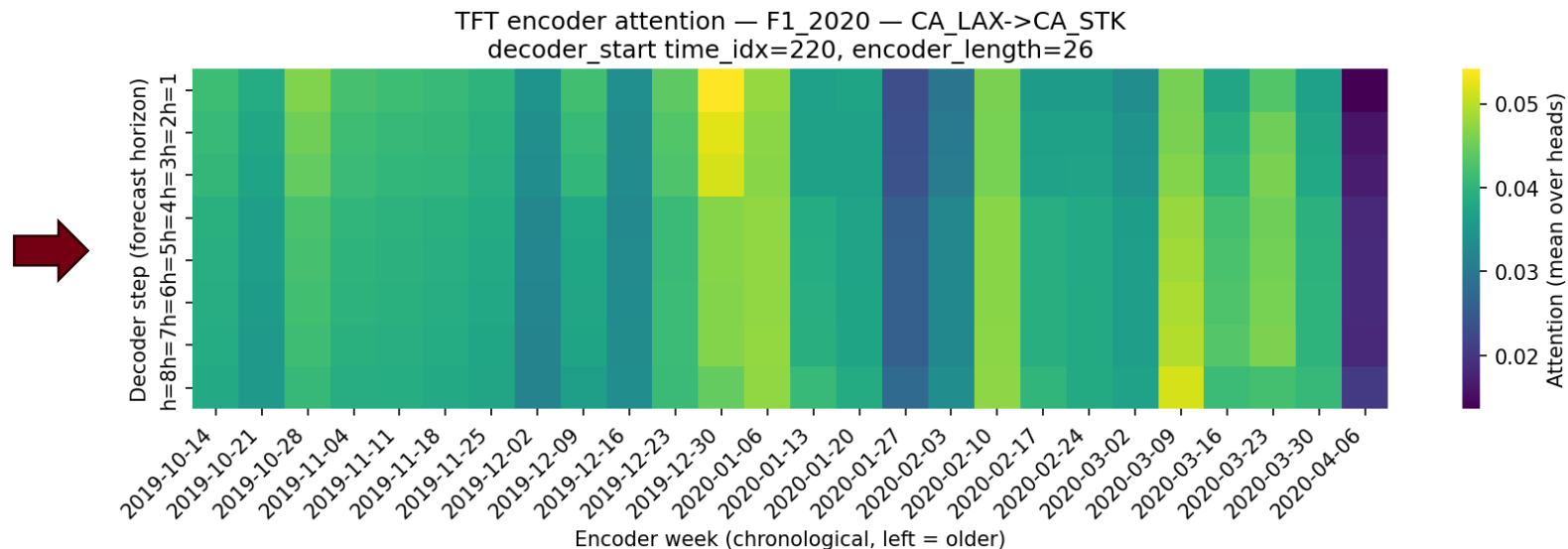
- **Geography and regime:** TFT is strongest on several Western / CA lanes in F1_2020 and F2_2021 (often across multiple horizons); LightGBM is more reliable on several Eastern / PA / GA lanes, especially when those lanes are orange across most of the grid.
- **Super-cycle vs calm:** In F2_2021, TFT still “owns” big blocks (notably CA lanes and some PA corridors), which matches the story that sequence + attention can help when dynamics are stressed or uncertain, but not everywhere.
- **2022 pullback:** In F3_2022, LightGBM dominates most cells; TFT only wins sparse (lane, horizon) pockets. That suggests a structural shift where tree + tabular signals generalize better under that year’s patterns (or where the TFT setup is less aligned with the data).
- **Horizon matters:** On lanes like TX_HOU→TX_DAL, short horizons often favor LightGBM while longer (e.g. 8-week) horizons sometimes favor TFT, so different planning horizons may warrant different models or ensembles, not a single rule.

As a learning exercise we tried to interpret the transformer's attention weights on CA_LAX --->> CA_STK

What are we looking at?

Each row is a decoder step (forecast lead: $h = 1 \dots 8$).
Each column is one encoder week (past context, 26 weeks ending just before the forecast starts). The color is mean attention over heads for that (horizon, past week): brighter = more weight on that history when forming that step's representation before the final quantile heads.

So this is: for this lane (CA_LAX→CA_STK), fold F1_2020, forecast origin with decoder_start time index 220, which past weeks the TFT's interpretable attention layer emphasizes, and how that changes by horizon.



1. Strong focus around late Dec 2019 / early Jan 2020

For $h = 1-4$, attention is concentrated on 2019-12-30 (and nearby weeks). In plain language: for short horizons, the model is leaning heavily on year-turn dynamics (holidays, Q4/Q1 behavior, etc.) rather than spreading weight evenly over all 26 weeks.

2. Another peak at early March 2020, strongest at $h = 8$

2020-03-09 lights up most for the longest horizon (bottom row). That often shows up when longer-horizon states need older or mid-history context (macro/seasonal/shock timing) while short horizons stay tied to very recent seasonality. Here it suggests the 8-week-ahead representation is “looking” further back into Q1 2020 than the 1-week-ahead path.

3. Very little weight on the last encoder week (2020-04-06)

Attention is cold on the most recent history for all h . That does not automatically mean “recent prices don’t matter”; the model can encode recency elsewhere.

4. Mid–late Jan / early Feb dip

Darker bands around late Jan – early Feb mean this layer down-weights those weeks relative to Dec–Jan and March for this window, again, conditional on the rest of the architecture.

Managerial Implications and Future Direction



Managerial Implications

- Forecasts must specify use case: point accuracy vs tail risk vs interval reliability (coverage).
- Domain features (e.g. macro drivers) may help interpretable models; validate out-of-sample and by lane / horizon.
- Don't default to deep learning when tabular + strong supervised loss already matches the decision problem.
- Inductive bias matters: tree ensembles align well with heterogeneous, tabular weekly panels; Transformers need scale, tuning, and often more regularization when n is modest.
- Quantile forecasts need calibration checks, not only average loss.
- Walk-forward / nested evaluation is what makes claims about generalization credible.

Future Direction:

- More hyperparameter search and capacity experiments for TFT; multi-seed stability.
- Per-lane error and regime shifts (e.g. stress folds); ablation of macro blocks.
- Ensembling or distillation if you want TFT's flexibility with GBM's reliability.

Thank you

Appendix

Results w/ Exogenous Data

